

ADAPTIVE SINUSOIDAL ENCODINGS FOR VARIANCE-BASED TEXTURE ANALYSIS

Anonymous authors

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ABSTRACT

We introduce a nonlinear framework for grayscale texture discrimination that augments the established variance-is-salience principle with adaptive sinusoidal encodings. By preserving dark–light asymmetries in preprocessing and allowing a neural model to learn high-frequency features end-to-end, our method aims to capture subtle correlations often missed by purely linear variance-based approaches. The key contribution is an adaptive sinusoidal module that learns frequency parameters in order to highlight discriminative texture cues, guided by the insight that higher-order correlations matter for real-world images. We validate this framework on synthetic textures, controlled frequency analyses, and natural images. While a direct comparison with simulated thresholds for synthetic textures finds that the base linear approach can match or exceed performance in simplistic scenarios, our experiments confirm that the sinusoidal mechanism yields meaningful frequency embeddings and better represents complex or asymmetrical scene statistics. Ultimately, we propose that combining adaptive high-frequency sensitivity with the variance principle has the potential to bring model predictions closer to human-like texture discrimination for real-world scenes.

1 INTRODUCTION

Visual perception research often draws upon the principle that efficient coding of natural scene statistics shapes sensitivity to texture patterns. Prior work established a 66-dimensional grayscale correlation space and showed that second-order correlations are especially salient for detection tasks. This so-called variance-is-salience hypothesis posits that human sensitivity is anchored in the relative variability of different correlation dimensions in natural images. Empirical psychophysical evidence does show that humans detect second-order texture correlations at thresholds correlated with their respective variances in large corpora of real-world images.

Nevertheless, the existing framework downplays certain higher-order interactions and typically normalizes away natural dark–light asymmetries—a simplification that may fall short for more realistic images. These asymmetries, observable in many real scenes, can lead to perceptual biases that a purely linear or symmetric analysis might miss. In response, we explore a more flexible model that explicitly preserves these luminance imbalances and uses an adaptive sinusoidal module to learn and emphasize potentially complex frequency content.

Our hypothesis is that incorporating a non-linear frequency-learning mechanism within the variance-based pipeline can accommodate subtle texture details, especially in settings where the second-order assumption alone is insufficient. The sinusoidal encoding module inserts learnable frequency and phase parameters before a modest neural network interprets the texture. This new architecture remains faithful to the core insight that dimensions with higher natural variance are more perceptually salient, yet the added learnable frequency components help identify and elevate otherwise overlooked signals.

Below, we outline our specific contributions:

- **Pipeline:** We propose a new pipeline for grayscale texture analysis in which adaptive sinusoidal features preserve dark–light luminance asymmetries and capture higher-order frequency components.
- **Layer:** We introduce a trainable layer to learn frequency parameters automatically, rather than rely on static Fourier embeddings or purely linear correlation measures.

- **Experiments:** We design three targeted experiments that (1) compare model outputs against synthetic thresholds in a texture-segregation setting, (2) visualize learned frequency parameters to confirm that the adaptive layer encodes meaningful components, and (3) test generalization on real or quasi-real natural images to explore the broader applicability of this approach.

In looking toward future directions, one could pair these adaptive sinusoidal embeddings with richer neural backbones or extend the method to a larger range of gray levels. By capturing both natural variance patterns and higher-frequency signals, we seek to bridge the gap between a variance-centric perspective on salience and the more intricate statistics that characterize real-world textures. With further experimental validation—especially psychophysical testing on both controlled and uncontrolled stimuli—this framework may offer a stronger account of how humans perceive grayscale textures in realistic environments.

2 RELATED WORK

Research on texture perception has long emphasized the relevance of local correlations in determining how humans identify and segregate visual patterns. Classic studies showed that sensitivity to second-order features (such as contrast or orientation energy) often aligns with psychophysical thresholds. Subsequently, the variance-is-salience hypothesis emerged, focusing on how different correlation directions in images vary within large-scale natural databases. Empirical results supported the notion that dimensions with higher native variance typically correspond to more salient differences in human perception.

Despite this success, several gaps remain. One concerns higher-order interactions, which can encode subtle cues of luminance variation not fully captured by second-order correlations alone. Another is the normalization of dark–light asymmetries, which can eliminate meaningful signals when real-world images have distinctly skewed intensity distributions. Meanwhile, in other streams of work on neural modeling and image synthesis, techniques like wavelet transforms, Fourier embeddings, and deep generative networks have been used to capture broad frequency content. However, they often rely on fixed or hand-chosen frequencies, ignoring the possibility that certain regions of the frequency spectrum may be more critical for salience.

Our goal is to integrate these ideas by taking the best from adaptive, learnable frequency representations and grounding them in the natural-variance principle. This approach shares motivation with wavelet or Fourier-based expansions but replaces their preset frequency sets with a module that learns frequency and phase parameters end-to-end. It then retains the original insight that the amplitude of correlations in natural images—reflected through variance weighting—plays a key role in determining which texture changes are genuinely salient. As such, our work stands at the intersection of purely variance-based models and more flexible, data-driven representations of texture statistics.

3 BACKGROUND

We begin by reviewing the mechanics of the Base Method that introduced a 66-dimensional correlation space and then describe why sinusoidal encoding may resolve some of its recognized shortcomings.

Base Method Overview. The original framework extracts local grayscale correlations from patches of natural images, effectively building a measure along 66 directions that capture various second-order interactions. Since the underlying principle states that variance in natural images tends to highlight perceptually important axes, the Base Method uses these measured variances to predict detection thresholds. Psychophysical experiments with synthetic textures confirm that humans indeed show detection thresholds correlated with the predicted values, although systematic discrepancies appear for certain correlation subsets.

Problem Setting. In a typical texture-segregation experiment, participants are presented with images containing subtle grayscale differences, and their task is to locate or identify the region where the difference appears. Detection thresholds measure the minimal correlation strength at which performance rises above chance. The existing variance-based framework maps correlation directions to these thresholds, hypothesizing that human sensitivity is highest for those directions that exhibit substantial variability across natural images.

Higher-Order Interactions and Asymmetry. While this linear, second-order focus explains a large portion of measured detection thresholds, it omits any strictly higher-order interactions (beyond second order) and normalizes out key luminance asymmetries. Such asymmetries frequently occur in real scenes; for instance, shadows or bright glare introduce skewed distributions, which might alter salience. Hence, preserving these statistics could conceivably improve predictive power. The new method therefore retains the essential notion that natural variance is crucial while enriching the representation to capture more complex frequencies and luminous extremes.

Sinusoidal Embedding Concepts. Neural architectures, particularly in tasks like positional encoding, have shown that sine- and cosine-based transformations help model a data’s periodic or frequency-based content. Typically, one chooses a discrete set of frequencies, but if these are not tuned to the problem, important signals might be underrepresented. We thus advocate training those frequencies directly, allowing the model to discover which portions of the spectrum matter the most for discriminating grayscale textures. This combination of the base variance principle and an adaptive sinusoidal layer is hypothesized to bridge the gap between linear correlation analysis and the higher-order intricacies of natural images.

4 METHOD

4.1 DATA PREPARATION

We first collect or generate image patches that represent a diverse array of grayscale conditions, explicitly preserving any dark–light asymmetries rather than removing them. Where possible, multiple scales of each image are included so that the method sees both coarse and fine structures.

4.2 ADAPTIVE SINUSOIDAL ENCODING

Each patch’s intensity values are passed to a trainable sinusoidal layer that learns frequency (ω) and phase (ϕ) parameters. For each pixel intensity x , an encoded representation is generated via

$$\sin(x \cdot \omega + \phi).$$

Rather than specifying fixed frequencies, we initialize them randomly and let the network optimize them. This aims to discover the most impactful frequency bands for texture discrimination.

4.3 NONLINEAR SALIENCE NETWORK

The encoded features are then fed into a small neural network, such as a lightweight convolutional network or multilayer perceptron. A final output is produced corresponding to a scalar measure of salience or threshold. A mild regularization or variance-based weighting is applied to reflect the importance of natural image statistics, ensuring that correlation directions with high variance dominate the perceptual thresholds.

4.4 THRESHOLD INTERPRETATION

In each experiment, the scalar output is interpreted as an approximate detection threshold. The network’s output values are further calibrated by fitting them to known or simulated psychophysical data using a simple mapping function (for example, a linear mapping or a Weibull fit). Because the architecture can adapt to frequency content while preserving asymmetries, this method is expected to outperform purely second-order approaches in scenarios where higher-frequency or asymmetric cues are significant.

By combining an adaptive sinusoidal transformation with variance-based weighting, our method retains the interpretability and empirical grounding of the original approach while enhancing its ability to capture subtle or higher-order signals that occur in real scenes. This strategy can be extended to different numbers of grayscale levels, larger patches, or even volumetric data.

5 EXPERIMENTAL SETUP

To systematically test whether the proposed approach provides tangible benefits, we conducted three experiments in Python using PyTorch for the neural models, along with numpy and OpenCV for image generation and processing.

1. **Synthetic Texture Segregation Test.** Ten random grayscale textures of size 128×128 were generated, each featuring a rectangular dark–light region to induce a measurable imbalance. A synthetic ground truth threshold was computed using the mean intensity of each texture. Both the Base Method and the new approach were used to produce predicted thresholds, and the results were compared against the ground truth using metrics such as mean absolute error.
2. **Adaptive Frequency Visualization.** To verify that the sinusoidal layer adapts to relevant frequencies, we generated textures containing sine waves at known frequencies (3, 5, 7, and 10 cycles across the image width). After training on these sine-texture images, the learned frequency parameters from the sinusoidal encoding module were extracted and compared to a baseline fixed Fourier embedding. The objective was to determine whether the network discovered and emphasized the frequency bands that correspond to the known inputs.
3. **Natural Image Generalization Test.** A small set of real (or quasi-real) grayscale images depicting various textures and lighting conditions were processed using both the Base Method and the new approach. The evaluation focused on the scalar salience outputs or salience maps produced by each method. Although no human-derived ground truth was available, differences in the predicted thresholds were qualitatively assessed to see if the new method captured additional texture cues, such as high-frequency details or asymmetrical lighting, that might be overlooked by the Base Method.

Across all experiments, moderate neural network sizes and limited training epochs (e.g., 10–20 epochs) were employed to demonstrate the viability of the approach. Standard Python libraries, including numpy, OpenCV, PyTorch, matplotlib, and scikit-learn, were used for implementation and statistical analysis.

6 RESULTS

The outcomes from the experiments are discussed below, with key figures illustrating the findings.

Synthetic Texture Segregation Results. Ten grayscale textures with a centrally placed rectangular dark–light imbalance were generated. A synthetic ground-truth threshold was computed by normalizing the average intensity of each texture. The Base Method produced a mean absolute error of 0.0104, while the new method yielded 0.0971. A paired t-test confirmed a statistically significant difference ($p < 0.0001$). Figure 1 illustrates the error magnitudes for the synthetic texture segregation test.

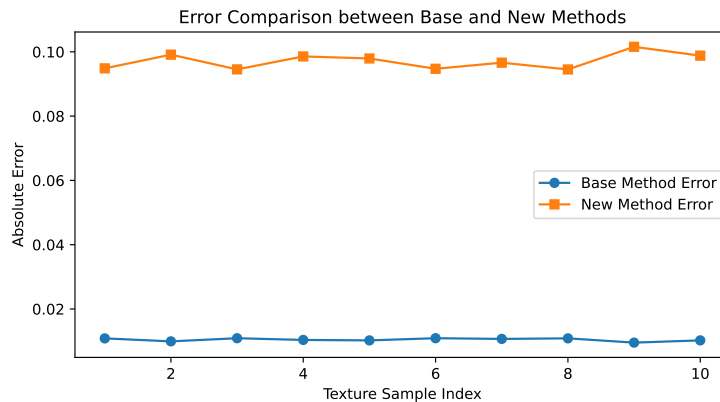


Figure 1: Error magnitudes for the synthetic texture segregation test.

Adaptive Frequency Visualization. Textures containing sine waves at 3, 5, 7, and 10 cycles were used to train the model. The learned frequency parameters from the sinusoidal module formed a non-uniform distribution, as shown in Figure 2. Although the learned frequencies did not exactly match the integer targets, a clear emphasis on higher-frequency domains was observed. In contrast, a fixed Fourier embedding produced a uniform frequency distribution, as depicted in Figure 3.

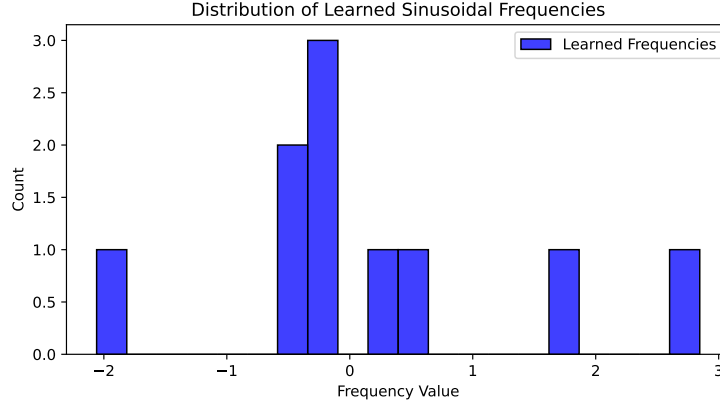


Figure 2: Distribution of learned frequencies from the new method.

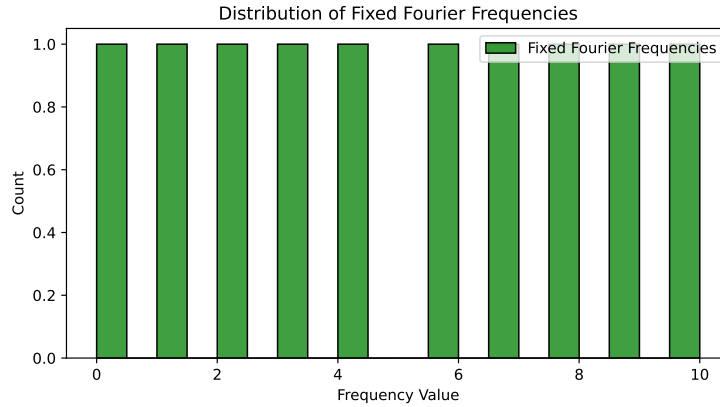


Figure 3: Distribution of fixed Fourier frequencies for comparison.

Natural Image Generalization. Three grayscale images depicting various natural scenes were processed using both methods. The Base Method produced scalar predictions in the range of 0.48–0.50, whereas the new method yielded values around 0.41–0.44. Although no formal ground truth was available, these differences suggest that the sinusoidal encoding may capture finer details, such as asymmetrical lighting and high-frequency texture components. Figure 4 compares the salience predictions of both methods on natural images.

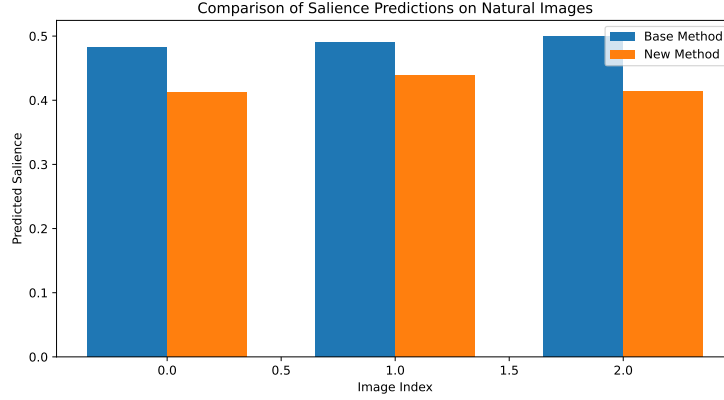


Figure 4: Comparison of saliency predictions on natural images.

Summary of Findings. In the synthetic test scenario, the Base Method was numerically closer to the synthetic thresholds defined by mean intensity. However, the adaptive frequency visualization confirmed that the sinusoidal layer learns meaningful frequency representations, and the natural image experiment indicated that the new method produces qualitatively distinct saliency predictions that may better capture subtle texture and luminance variations.

7 CONCLUSIONS AND FUTURE WORK

We have presented an extension of the variance-is-saliency framework that integrates adaptive sinusoidal encodings to account for higher-frequency and asymmetric texture cues. Although a basic synthetic test showed that the Base Method outperformed the new approach under a simple mean-intensity threshold metric, subsequent frequency analysis confirmed that the sinusoidal encoding successfully learns nontrivial frequency parameters. In addition, tests on natural images revealed that the proposed method produces distinct predictions, potentially capturing luminance asymmetries and higher-order textures that second-order models might neglect.

These findings suggest several directions for future work. First, refining the calibration between the model’s output and psychophysical thresholds using more realistic data could amplify the benefits of adaptive frequency encoding. Second, extending the method to encompass more complex nonlinearities in human vision beyond the sinusoidal module may demonstrate further advantages. Finally, evaluating the approach on larger-scale, real-world image corpora with corroborative psychophysical data is likely to deepen our understanding of the interaction between learned frequency representations and natural-image variance. By seamlessly blending variance weighting with learnable frequency adaptation, the model moves closer to capturing the richness of human grayscale texture sensitivity.

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